

Roll No. U121CS43



Indian Institute of Information Technology, Surat

Department of Computer Science Engineering

End-Semester Examination, 7th December- 2023

Fuzzy Logic and Neural Networks (CS 521)

B.Tech – 3rd Year (Semester-5th)

Time: 10:00 AM – 01:00 PM

Marks: [100]

Note: 1. Attempt all the questions. 2. Make a necessary assumptions wherever required.

Q.1 [A] We want to compare two sensors based upon their detection levels and gains setting [10]
for a universe of discourse of gain settings $X = \{0, 20, 40, 60, 80, 100\}$, the sensor detection levels for the monitoring of a standard item provides typical membership functions to represent the detection levels for each of the sensors; these are given below in standard discrete form

$$S1 = 0.0/0 + 0.5/20 + 0.65/40 + 0.85/60 + 1.0/80 + 1.0/100$$

$$S2 = 0.0/0 + 0.45/20 + 0.6/40 + 0.8/60 + 0.95/80 + 1.0/100$$

Find the following membership functions using standard fuzzy operations

(1) $\mu_{S1 \cup S2}(x)$ (2) $\mu_{S1 \cap S2}(x)$ (3) $\mu_{\overline{S1} \cup S1}(x)$ (4) $\mu_{\overline{S1} \cap S2}(x)$

(5) α -cut for each set where $\alpha = 0.5$, and $\alpha = 0.3$

[B] In the field of computer networking there is an imprecise relationship between the [10]
level of use of a network communication bandwidth and the latency experienced in peer-to-peer communications. Let X be the fuzzy set of use levels (in terms of percentage of full bandwidth used) and Y be a fuzzy set of latencies (in milliseconds) with the following membership functions:

$$X = 0.2/10 + 0.5/20 + 0.8/40 + 1.0/60 + 0.6/80 + 0.1/100$$

$$Y = 0.3/0.5 + 0.6/1 + 0.9/1.5 + 1.0/4 + 0.6/8 + 0.3/20$$

Find the Cartesian product represented by $R = X \times Y$.

Now, suppose we have a second fuzzy set of bandwidth usage given by

$$Z = 0.3/10 + 0.6/20 + 0.7/40 + 0.9/60 + 1/80 + 0.5/100$$

Find $S = Z \circ R$ using max-min composition.

Q-2 [A] Attempt Any One Question

[05]

1) Develop the fuzzy membership functions for each of the following:

(i) Person X is considerably heavier than 100 pounds

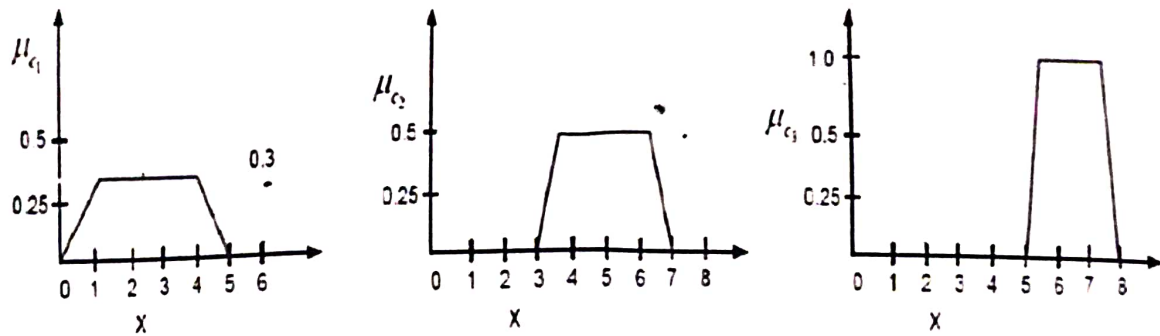
(ii) Car Z costs approximately \$30,000

(iii) $\mu_A(x) = 0$ for $x \leq 5$ and $1 - (x - 5)$ when $x > 5$

✓ 2) Develop a reasonable membership function for the following fuzzy sets based on height measured in centimeters: (i) "tall" (ii) "short" (iii) "not short"

Q.2 [B]

[15]



Find the crisp output from the union of fuzzified outputs by using any one of the defuzzification methods in detail with theory.

- Q-3 [A]** Explain how you would design a washing machine to have fuzzy inference in a real world environment. In your discussion include the following factors: [10]
- (i) A practical choice of the fuzzy input & output variables
 - (ii) A practical no. of quantizations and choice of linguistic labels
 - (iii) Draw and indicate a reasonable membership functions for each fuzzy variable.
 - (iv) Indicate practical values of universe of discourse
 - (v) Three examples of rules
 - (vi) The most practical approach of defuzzification technique
- [B]** Explain the operation of dendrite, soma, and axon in the biological neuron with diagram. [07]
- [C]** What are the basic conditions that the fuzzy expression should satisfy to be a fuzzy logic? [03]
- Q.4 [A]** Define linguistic variable? What are the fuzzy linguistic variable basic parts? Define components for the linguistic variable X whose name is temperature? [05]
- [B]** Explain any two learning methods available in neural networks in detail. [05]
- [C]** Let R be a fuzzy relation defined on the set of cities representing the concept 'very near'. Discuss the properties of fuzzy equivalence relation. [05]
- [D]** Explain Fuzzy C-Means Algorithm in detail. [05]
- Q.5 [A]** Explain different types of Neural Networks in detail. [05]
- [B]** List out the different applications of Fuzzy Logic in detail. [05]
- [C]** Explain the relationship between Neural Network and Artificial Intelligence in detail. [05]
- [D]** Explain any two Automated Methods for Fuzzy Systems. [05]